Addressing Train-Serve Skew in Solar Irradiance Forecasting in Northern Vietnam: A Robustness and Calibration Framework

Nguyen Huong Thao *, Nguyen Thanh Nam *, Nguyen Thi Anh Tho *, Bui Hanh Trang, Tran Thi Thanh Hai
School of Electrical and Electronic Engineering
Hanoi University of Science and Technology
Hanoi, Vietnam
Email: *nam.nt233870@sis.hust.edu.vn*

***Abstract*—Accurate Global Horizontal Irradiance (GHI) forecasting is critical for solar power integration. However, current machine learning models often suffer from "Train-Serve Skew"due to their reliance on imperfect weather forecasts during actual operation. Furthermore, traditional point forecasting fails to quantify uncertainty, limiting its value for grid risk management. To address these challenges, this paper proposes a proactive two-phase probabilistic forecasting framework using the Clearness Index (K_t). Phase 1 evaluates the robustness of base models by simulating weather forecast errors via Monte Carlo noise injection. Phase 2 applies Conformal Prediction to integrate both weather data uncertainty and internal model uncertainty, generating reliable prediction intervals. The framework is empirically validated using various machine learning algorithms across three distinct microclimates in Northern Vietnam. Results reveal significant regional variations in sensitivity to weather noise; for instance, under high noise conditions, the Robustness Score drops severely to 0.69 in the most vulnerable region, whereas the most resilient region maintains a score of 0.78. Crucially, the proposed framework successfully restores system reliability, raising the Prediction Interval Coverage Probability (PICP) to ideal levels exceeding 85% across all noise scenarios. Ultimately, this approach transforms forecast uncertainty into quantifiable risk warnings, providing grid operators with a robust tool for safe and optimal power dispatch decisions.**

***Index Terms*—Global Horizontal Irradiance Forecasting, Forecast Uncertainty, Robustness Assessment, Monte Carlo Simulation, Prediction Intervals, Conformal Prediction, Solar Energy Forecasting.**

I. Introduction

Solar energy plays a crucial role in the global energy transition, with Global Horizontal Irradiance (GHI) directly determining the power output of photovoltaic systems. Recently, machine learning (ML) models have proven highly effective in forecasting GHI by capturing complex non-linear relationships. Accurate forecasting is especially urgent in Northern Vietnam, a key area for distributed renewable energy that currently faces power shortage risks. However, this region features a highly complex microclimate characterized by four distinct seasons, urban fog, and sudden thunderstorms. Therefore, a forecasting system must provide not only high accuracy but also evaluate the algorithm's robustness and forecast errors. This provides the essential scientific foundation for investors and grid operators to manage risks, calculate opportunity costs, optimize battery energy storage systems, and ensure safe grid planning. Despite achieving high accuracy in laboratory settings, current ML models often suffer from "Train-Serve Skew."This issue arises because models are trained on perfect historical observation data but must rely on imperfect weather forecasts during actual operation. While recent studies [1], [2], [3] have successfully improved accuracy, they face a major research gap: they do not evaluate model performance degradation under weather forecast uncertainty. Furthermore, most current systems are limited to point forecasting, which completely hides operational risks. Even though some studies [4] have transitioned to probabilistic forecasting, they still struggle to simultaneously integrate the uncertainty from the input weather data and the internal uncertainty of the model itself. To fully address these limitations, this study proposes a proactive two-phase framework that transitions from stress-testing algorithms to establishing safe prediction intervals. The first phase directly evaluates the robustness of ML models against weather forecast errors using Monte Carlo perturbation techniques. Subsequently, the second phase applies the Conformal Prediction algorithm to construct highly reliable prediction intervals, successfully combining both weather forecast uncertainty and model uncertainty. This framework is then empirically validated across three distinct microclimates in Northern Vietnam (Hanoi, Bac Giang, and Son La), providing deep insights and practical tools for local energy dispatch strategies.

II. Related works

A. Robustness under Weather Forecast Uncertainty

In machine learning, "Train-Serve Skew"occurs when the operational data distribution differs from the training data. [5] highlighted that ignoring this shift creates "hidden technical debt",leading to severe performance drops in real-world deployments. In solar forecasting, models trained on perfect

*These authors contributed equally to this work.

Table I: Comparison of Methodologies Regarding Weather Uncertainty

Ref.	Methodology	Target	Addressed Weather Uncertainty?	Limitations / Gaps
[1]	Benchmarking ML models	K_t	No	Assumes perfect weather inputs.
[2]	ARIMA Time-Series	K_t, K_c	No	Fails in highly volatile climates.
[3]	K-means + SVR	K_t	No	Assumes perfect weather inputs.
[4]	SVR + Laplace Distribution	GHI	Partial (Reactive)	Analyzes residuals only; no proactive stress-testing.
Ours	**Monte Carlo + Conformal Prediction**	K_t	**Yes (Proactive)**	**Addresses Train-Serve Skew directly; adaptable to complex climates.**

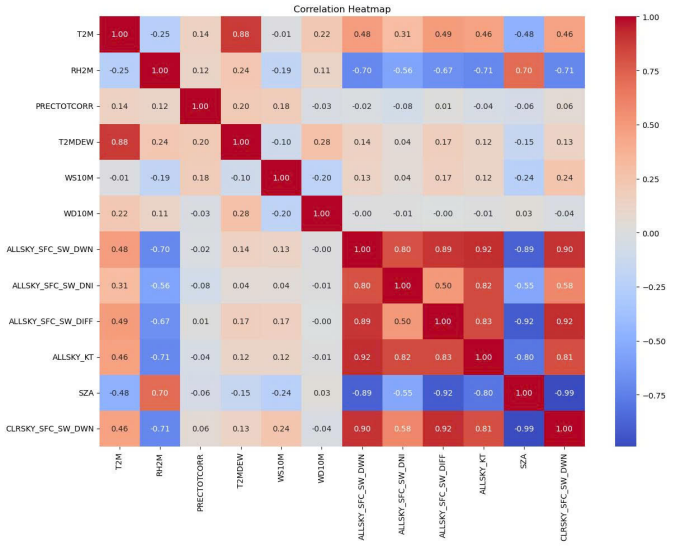


Figure 1: Correlation Matrix

historical weather data often struggle when fed with imperfect weather forecasts during operation. However, few studies directly evaluate model robustness by simulating these weather forecast errors.

B. Target Transformation via Clearness Index

Predicting Global Horizontal Irradiance (GHI) directly is challenging due to daily and seasonal trends. [1] demonstrated that transforming the target variable to the Clearness Index (K_t) removes these trends and significantly improves non-linear machine learning models. Following this approach, [2] used ARIMA models to predict K_t in Algeria. While highly accurate in stable desert climates, the model's performance dropped sharply in volatile regions. Alternatively, [3] combined K-means clustering with Support Vector Regression (SVR) to predict K_t, successfully reducing forecast errors compared to standard SVR models.

C. Probabilistic Solar Forecasting

Grid operators require interval forecasting for safe decision-making. [4] emphasized the importance of combining point and interval forecasts. They achieved high point-forecast accuracy using SVR, and subsequently fitted the residual errors to a Laplace distribution. This allowed them to generate confidence intervals and quantify forecast uncertainty.

D. Research Gap

A common limitation in the aforementioned studies is the assumption of perfect weather input data. By overlooking Train-Serve Skew, these models risk severe performance degradation in actual operation. Furthermore, current uncertainty analyses [4] are primarily "reactive"—they analyze residuals only after the model runs in ideal conditions. There is a clear need for a "proactive"framework that stress-tests models against simulated weather forecast errors and robustly calibrates prediction intervals, especially for complex climates like Northern Vietnam. To summarize the discussed literature and highlight the motivation for our proposed method, Table I presents an overview of the related works and their respective limitations.

III. Proposed Methods

The proposed two-phase framework for GHI forecasting robustness evaluation and probabilistic uncertainty quantification is illustrated in Fig. 2. In the following, we will describe the methodology underlying each phase.

A. Dataset and Preprocessing

Hourly meteorological and solar radiation data from January 1, 2020, to December 31, 2025, were collected from the NASA POWER database [6] for three representative Northern Vietnam locations: Hanoi, Bac Giang, and Son La. The dataset contains 12 physical variables describing time, meteorology, astronomy, and solar radiation conditions.

To isolate stochastic atmospheric attenuation from deterministic astronomical variations, the forecasting target was transformed from GHI to the Clearness Index, defined as

$$K_t = \frac{GHI}{H_0} \tag{1}$$

where H_0 is the extraterrestrial horizontal irradiance [1], [7]. During inference, the predicted GHI is simply reconstructed as

$$\widehat{GHI} = \widehat{K_t} \times H_0 \tag{2}$$

To rigorously determine the most relevant predictors and ensure a data-driven feature selection process, an Exploratory Data Analysis (EDA) utilizing a Pearson correlation heatmap was conducted (Fig. 1). Based on the correlation analysis, several variables were systematically excluded from the input set:

- **Target Leakage Prevention:** Direct Normal Irradiance (ALLSKY_SFC_SW_DNI) and Diffuse Horizontal

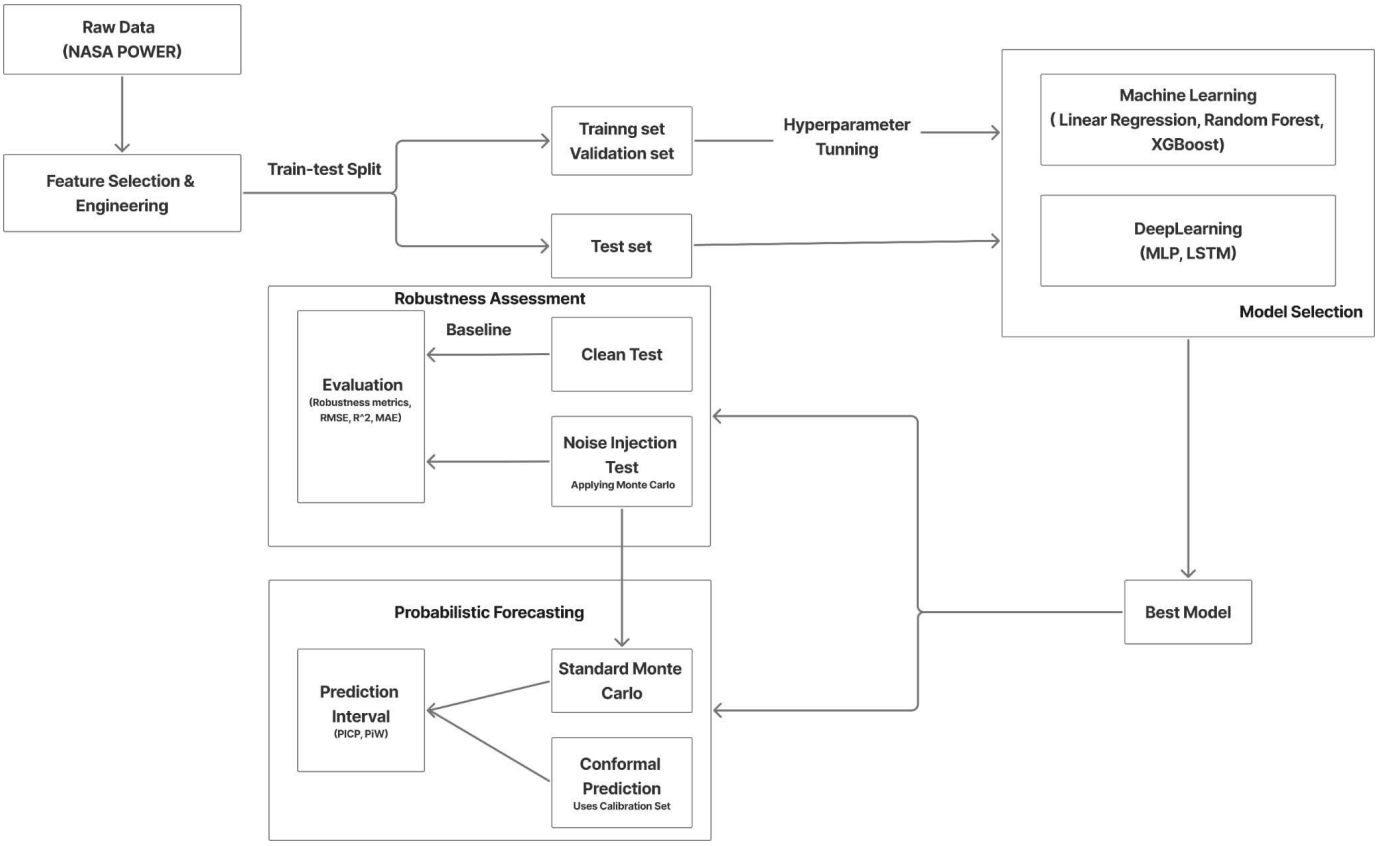


Figure 2: Overall of the proposed method

Irradiance (ALLSKY_SFC_SW_DIFF) exhibit exceptionally high positive correlations with both the Clearness Index ($r = 0.82$ and $r = 0.83$, respectively) and the original GHI ($r = 0.80$ and $r = 0.89$, respectively). Retaining these direct physical components would introduce severe target leakage into the model; hence, they were completely excluded [8].

- **Multicollinearity Reduction:** Dew/Frost Point Temperature (T2MDEW) demonstrates a strong multicollinear relationship with Air Temperature (T2M) at $r = 0.88$. Since the dew point is inherently mathematically derived from temperature and relative humidity, T2MDEW was dropped to reduce dimensionality and redundancy, while retaining T2M and RH2M.
- **Irrelevance:** Wind Direction at 10m (WD10M) shows virtually zero correlation with both GHI ($r = -0.00$) and K_t ($r = -0.01$). Given its lack of statistical significance and physical relevance to solar radiation, it was removed from the predictor set.

For data preprocessing, temporal variables were cyclically encoded using sine and cosine transformations, and all numerical features were standardized using Z-score normalization fitted strictly on the training set. Non-informative nighttime samples (Solar Zenith Angle $\geq 90°$) were removed, retaining approximately 27,500 daylight observations per site. This filtering step eliminates non-informative observations and restricts the forecasting task to daylight conditions, where solar irradiance prediction is meaningful [2]. Finally, to strictly preserve chronological order and avoid temporal data leakage, the dataset was partitioned into a training set (Jan 2020 – Feb 2025), a validation set for hyperparameter tuning (Mar 2025 – Aug 2025), and an independent test set for final evaluation (Sep 2025 – Dec 2025). The final 11 selected predictors are summarized in Table II.

Table II: Final Predictor Set

Category	**Features**
Temporal	Hour (sin, cos), Month (sin, cos)
Meteorological	Temperature (T2M), Humidity (RH2M), Precipitation (PRECTOTCORR), Wind Speed (WS10M)
Astronomical	Solar Zenith Angle (SZA), Extraterrestrial Horizontal Irradiance (H_0), Clearsky Global Horizontal Irradiance (CLRSKY_SFC_SW_DWN)
Prediction Target	Clearness Index (K_t)

B. Forecasting Models

Five forecasting models representing different machine learning paradigms were investigated, including Ridge Regression, Random Forest (RF), Extreme Gradient Boosting (XGBoost), Multilayer Perceptron (MLP), and Long Short-

Term Memory (LSTM). These models were selected to provide a diverse comparison between linear regression, tree-based ensemble learning, and deep neural network approaches.

For each study area, every candidate model was trained using the training set. Model-specific hyperparameters were optimized on the validation set, and the optimal configuration for each model was retained. Consequently, five optimized forecasting models were obtained for each study area.

The optimized models were subsequently evaluated once on the independent test set using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination R^2. The purpose of this evaluation was not to further optimize the models, but rather to identify the strongest forecasting baseline for each climatic region. The model achieving the best predictive performance on the test set was then fixed and used throughout all subsequent experiments. No additional training, hyperparameter tuning, or model selection was performed after this stage.

Since the objective of this study is to investigate model robustness under weather forecast uncertainty rather than to compare forecasting algorithms, both Phase I (Robustness Evaluation) and Phase II (Probabilistic Forecasting) were conducted exclusively using the selected baseline model for each study area. This experimental protocol ensures that all subsequent uncertainty analyses are performed on the most accurate forecasting model available for each climatic region while avoiding any further parameter optimization after the baseline model has been established.

C. Phase 1: Robustness Evaluation

Phase 1 assesses the baseline model's robustness against operational Numerical Weather Prediction (NWP) errors. We employ a Monte Carlo framework to inject targeted Gaussian noise into four meteorological variables—temperature (T2M), wind speed (WS10M), relative humidity (RH2M), and precipitation (PRECTOTCORR) —assigning higher perturbation probabilities to highly uncertain variables based on two complementary criteria: (1) exploratory data analysis (EDA), including statistical measures, which characterize the variability of each meteorological variable; and (2) meteorological knowledge regarding the predictability of different weather factors.

To investigate different levels of weather forecast uncertainty, three perturbation scenarios are defined, namely Low, Medium, and High. For each selected meteorological feature, Gaussian noise is injected according to

$$x_i' = x_i + \epsilon_i \tag{3}$$

where $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2)$ and the perturbation magnitude is determined by $\sigma_i = \lambda \cdot std(x_i)$

where $std(x_i)$ denotes the standard deviation of the (i)-th meteorological feature, and λ controls the perturbation intensity. In this study, three values of λ are considered to represent different uncertainty levels: Low $\lambda = 0.25$, Medium $\lambda = 0.50$, and High $\lambda = 1.00$. To realistically emulate the stochastic nature of weather forecast errors, a N Monte Carlo realizations simulation is employed. These perturbed weather scenarios are ealuated by the best-performing forecasting model selected in the previous stage. The overall performance metrics (RMSE, MAE, and R^2) are averaged across all N simulations.

To provide a normalized measure of robustness, a Robustness Score (RS) is further defined as

$$RS = 1 - \frac{RMSE_{mean} - RMSE_{clean}}{RMSE_{clean}} \tag{4}$$

where $RMSE_{clean}$ denotes the prediction error obtained on the clean, unperturbed testing dataset. An RS value close to 1 indicates minimal performance degradation under weather forecast uncertainty, whereas lower values indicate greater sensitivity to perturbations. Besides deterministic evaluation metrics, the empirical coverage of the Monte Carlo prediction intervals is also assessed using the Phase-I Prediction Interval Coverage Probability (PICP-P1), defined as

$$\frac{1}{M}\sum_{i=1}^{M} \mathbf{1}\left(y_i \in PI_i^{MC}\right) \tag{5}$$

where M is the number of testing samples, PI_i^{MC} denotes the prediction interval generated directly from the Monte Carlo prediction distribution for the (i)-th sample, and $\mathbf{1}\cdot$ is the indicator function. The PICP-P1 metric measures the proportion of observed GHI values contained within the Monte Carlo prediction intervals before conformal calibration. It therefore serves as the baseline coverage metric for comparison with the calibrated Prediction Interval Coverage Probability (PICP-P2) obtained in Phase II. The Monte Carlo prediction distributions together with their corresponding prediction intervals are subsequently propagated to Phase 2, where Split Conformal Prediction is employed to further calibrate the intervals and improve their empirical coverage.

D. Phase 2: Probabilistic Forecasting

1) Motivation and Rationale: In practical power grid operations, point forecasts are insufficient for effective risk management. While accounting for weather forecast errors (Forecast Uncertainty) is a necessary first step, it does not fully reflect the total uncertainty of the operational environment. This partial approach ignores a critical factor: Model Uncertainty. Even when provided with perfect input data, machine learning algorithms inherently produce errors due to their structural limitations. Therefore, the primary motivation of Phase 2 is to construct highly reliable prediction intervals that capture the complete spectrum of uncertainty. By integrating both weather-induced and model-induced errors, this phase aims to generate comprehensive probabilistic forecasts. These robust intervals are essential to provide grid operators with the accurate risk assessments needed to make safe and effective dispatch decisions in real-world scenarios.

2) Localized Model Uncertainty via K-Nearest Neighbours: In GHI forecasting, model errors are not uniformly distributed but change depending on the time of day (heteroscedasticity). For example, solar irradiance fluctuates much more at noon

than in the early morning or late afternoon. To address this, we apply Locally Weighted Conformal Prediction (LWCP) combined with the K-Nearest Neighbors (KNN) algorithm.

- Step 1: Calibration Set Storage. We collect all the prediction errors (residuals) from the best model on the clean Validation set: $\mathcal{E}_{\text{val}} = \{\epsilon_1, \epsilon_2, \ldots, \epsilon_{N_{\text{val}}}\}$, where $\epsilon_i = y_i - \hat{y}_i$.
- Step 2: Finding Similar Conditions. For each new data point in the Test set, we use KNN (with Euclidean distance) to find the K most similar past instances from the Validation set. This similarity is based on cyclic time features ($\text{hour}_{\sin}$, $\text{hour}_{\cos}$). This ensures the prediction interval is built using the actual error behavior from similar times of the day.
- Step 3: Quantile Extraction. From the K residuals of these nearest neighbors (denoted as $\mathcal{E}_K$), we empirically calculate the lower bound (q_{lo}) and upper bound (q_{hi}) of the base prediction interval at a significance level α (with $\alpha = 0.1$):

$$q_{\text{lo}} = \text{Quantile}(\mathcal{E}_K, \alpha/2)$$
$$q_{\text{hi}} = \text{Quantile}(\mathcal{E}_K, 1 - \alpha/2)$$

3) Integration of Forecast Uncertainty: The Total Uncertainty of the system is a combination of the model's internal uncertainty (calculated above) and the input data uncertainty. Let σ_{MC} be the standard deviation of the predictions obtained from N Monte Carlo noise injections for a specific data sample (the result from Phase 1). The final prediction interval C_{Xtest} is created by expanding the local CP bounds using σ_{MC}:

$$C_{X_{\text{test}}} = [y_{\text{test}} + q_{\text{lo}} - \sigma_{\text{MC}}, \quad y_{\text{test}} + q_{\text{hi}} + \sigma_{\text{MC}}] \tag{6}$$

This additive integration allows the system to be highly adaptive. When the weather data contains more noise, ${}_MC$ increases. Consequently, the prediction interval automatically widens to maintain safety and reliability.

4) Evaluation Framework: The quality of the prediction intervals is evaluated using two core metrics. These metrics balance operational safety with practical usefulness:

- **Prediction Interval Coverage Probability (PICP):** This metric evaluates reliability. We assume that the coverage from Phase 1 (PICP-P1) will be low because it does not include model uncertainty. Applying our proposed framework in Phase 2 should significantly improve this metric (PICP-P2), bringing the coverage close to the expected theoretical level of (1).
- **Prediction Interval Width (PIW):** This metric measures the average width of the prediction intervals. Let U_i and L_i denote the upper and lower bounds for the i-th sample, respectively:

$$\text{PIW} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (U_i - L_i) \tag{7}$$

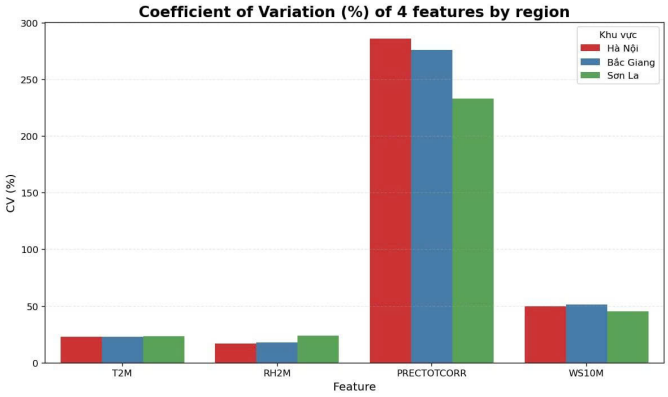


Figure 3: Coefficient of Variation of 4 weather features by region

While improving PICP-P2 is the primary objective for operational safety, the PIW must also be controlled. Excessively wide intervals reduce informative value for power grid dispatch decisions. Achieving a balance between PICP and PIW is the main criterion for verifying the effectiveness of the proposed Locally Weighted Conformal Prediction (LWCP) method.

IV. Experiment

1) Feature Analysis: To accurately simulate the impact of weather forecast errors in Phase 1, we first need to analyze the distribution of the input variables. The Coefficient of Variation (CV) chart for the four weather features across the three regions (Fig.3) shows clear differences. Precipitation (PRECTOTCORR) has extreme variability, with a CV exceeding 25% in all three areas. Wind speed (WS10M) shows moderate variability (around 50%). Meanwhile, temperature (T2M) and relative humidity (RH2M) are much more stable, maintaining very low CV levels (under 25%).

Although the CV reflects the natural variability of the observed data, the actual uncertainty of Numerical Weather Prediction (NWP) models depends on the physical nature of each variable. Based on meteorological evaluations, such as the high-resolution data assimilation study in Lai Chau, Vietnam [9], forecast uncertainty ranks hierarchically. Temperature (T2M) is highly stable and easiest to predict. Wind speed (WS10M) exhibits higher errors due to terrain friction but follows predictable large-scale patterns. Relative humidity (RH2M), despite its low natural variability, is non-linear and highly sensitive to cloud microphysics, making its forecast harder to control than wind speed [10]. Finally, precipitation (PRECTOTCORR) is highly localized and sudden, consistently yielding the highest forecast errors.

Consequently, applying uniform random noise across all variables is unrealistic. Instead, our Monte Carlo noise injection incorporates this exact uncertainty ranking: Temperature < Wind Speed < Relative Humidity < Precipitation. This approach ensures that our robustness testing accurately reflects the real-world forecast errors faced by solar power dispatch centers.

Table III: Optimal Hyperparameters for Selected Models across Microclimate Regions

Region	Selected Model	Optimal Hyperparameters
Hà Nội	MLP	'learning_rate_init': 0.0005, 'hidden_layer_sizes': (256, 128, 64), 'alpha': 0.01
Bắc Giang	XGBoost	'subsample': 0.8, 'n_estimators': 600, 'min_child_weight': 3, 'max_depth': 9, 'learning_rate': 0.01, 'gamma': 0.01, 'colsample_bytree': 0.9
Sơn La	LSTM	'units': [64, 32], 'dropout': 0.2, 'batch': 256, windowsize = 12

Table IV: Robustness Evaluation of Base Models Under Varying Weather Forecast Noise Levels

Region	Noise Level	RMSE	MAE	R^2	Robustness Score
Hà Nội	Clean	83.85	56.91	0.87	1.00
	Low	86.55	58.93	0.86	0.97
	Medium	91.68	62.24	0.84	0.91
	High	102.09	68.36	0.80	0.78
Sơn La	Clean	73.30	47.51	0.90	1.00
	Low	84.44	59.65	0.86	0.85
	Medium	85.57	60.52	0.86	0.83
	High	88.55	62.67	0.85	0.79
Bắc Giang	Clean	77.85	56.38	0.88	1.00
	Low	91.30	63.33	0.84	0.83
	Medium	95.73	66.02	0.82	0.77
	High	102.18	69.91	0.79	0.69

A. Experimental setting

To ensure a fair comparison, models were independently optimized for each region via RandomizedSearchCV (50 iterations) to minimize the validation RMSE. A PredefinedSplit cross-validation strictly preserved chronological order, preventing temporal data leakage. The optimal hyperparameters for the selected baseline models are detailed in Table III.

For the robustness evaluation, 50 independent Monte Carlo trials were conducted per location for each noise severity level ($\lambda \in \{0.25, 0.50, 1.00\}$). To realistically reflect the intermittent nature of NWP errors, a timestep perturbation rate of $p_t = 0.50$ was applied.

At each selected timestep, noise was injected per feature according to the perturbation probabilities as follows: PRECTOTCORR (0.8), WS10M (0.5), RH2M (0.4), and T2M (0.3), derived from the CV-based variability ranking in Section IV-A, under three severity levels: Low ($k = 0.25$), Medium ($k = 0.50$), and High ($k = 1.00$).

Locally Weighted Conformal Prediction was applied with a miscoverage level $\alpha = 0.10$ (targeting 90% nominal coverage). Each test sample's prediction interval is constructed from the empirical residual distribution of its $K = 25$ nearest neighbours in the validation set, conditioned on the pair ($hour_sin$, $hour_cos$)—a cyclic encoding of the hour of day.

V. Results and Discussion

A. Performance of Base Models

The evaluation of the base models on the clean test set shows that no single algorithm is optimal for all locations. The local geography and microclimate determine the most suitable model. In Hanoi, the Multilayer Perceptron (MLP) achieves the best performance with an RMSE of 83.85 and an R^2 of 0.87. In Son La, the Long Short-Term Memory (LSTM) model outperforms others, recording an RMSE of 73.30 and an R^2 of 0.90. Meanwhile, in Bac Giang, the XGBoost model is the most effective, yielding an RMSE of 77.85 and an R^2 of 0.88.

B. Phase I: Robustness Evaluation

Table V presents the Phase 1 evaluation results under different noise levels. As the simulated weather forecast noise increases (from Clean to Low, Medium, and High), the performance of all models drops significantly. Specifically, the RMSE increases while the R^2 decreases across all regions. For instance, the RMSE of the XGBoost model in Bac Giang rises sharply from 77.85 under clean conditions to 102.18 under high noise, while its R^2 drops from 0.88 to 0.79.

Comparing the regions reveals different levels of climate sensitivity. Hanoi demonstrates the highest robustness, maintaining a Robustness Score of 0.78 even at the high noise level. In contrast, Bac Giang is the most vulnerable region, with its Robustness Score falling steeply to 0.69 under high noise.

To explain this vulnerability, we analyze the SHAP (SHapley Additive exPlanations) values. Figure 4 and Figure 5 show the SHAP summary plots for the MLP model in Hanoi and the XGBoost model in Bac Giang, respectively. In Hanoi (Figure 4), the model relies heavily on stable astronomical variables such as the clear-sky irradiance and solar zenith angle (SZA). Conversely, in Bac Giang (Figure 5, dynamic weather variables like precipitation (PRECTOTCORR) and temperature (T2M) have a dominant impact on the output. This heavy reliance on volatile weather features explains why the Bac Giang model suffers the most severe performance degradation when these inputs are corrupted by noise.

C. Phase II: Probabilistic Forecasting

Table VI summarizes the results of the probabilistic forecasting in Phase 2. A key takeaway is the dramatic improvement in the Prediction Interval Coverage Probability (PICP). If the system only propagates input noise without calibrating the model (PICP-P1), the coverage is dangerously low. For example, the coverage for Son La is only 0.07 (7%) under low noise. However, after applying Conformal Prediction (PICP-P2), the coverage jumps to ideal levels, exceeding 0.85 (85%) across all scenarios. This successfully resolves the coverage deficit.

This high reliability comes with a necessary trade-off in the Prediction Interval Width (PIW). To maintain a PICP above 85% as weather noise increases, the algorithm must automatically widen the prediction interval. For example, the

Table V: Probabilistic Forecasting Performance Before and After Conformal Prediction Calibration

Location	Noise Level	PICP-P1	PICP-P2	PIW-P2
Hà Nội	Clean	–	0.850	251.0
	Low	0.232	0.896	282.8
	Medium	0.380	0.920	308.5
	High	0.531	0.941	343.9
Sơn La	Clean	–	0.851	204.6
	Low	0.070	0.887	218.9
	Medium	0.130	0.903	232.0
	High	0.239	0.924	254.6
Bắc Giang	Clean	–	0.849	231.8
	Low	0.341	0.948	296.9
	Medium	0.420	0.958	312.0
	High	0.494	0.964	330.1

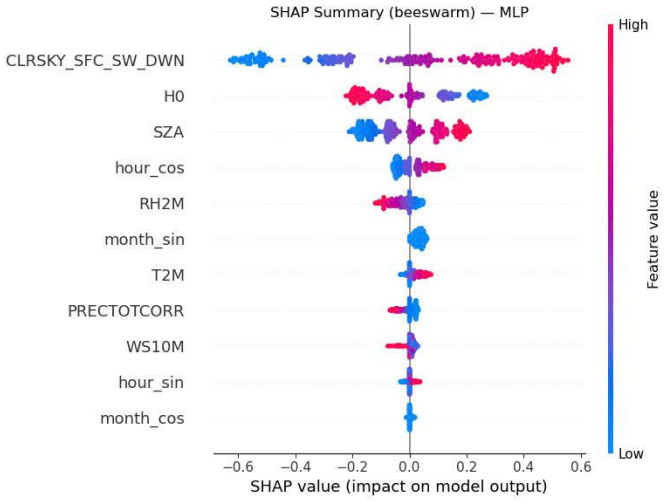


Figure 4: SHAP for Hà Nội

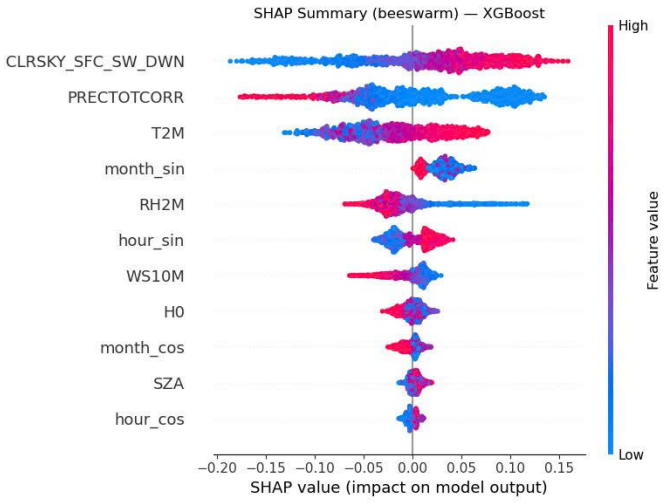


Figure 5: SHAP for Bắc Giang

PIW in Bac Giang expands from 231.8 under clean conditions to 330.1 under high noise.

VI. Conclusion

This study addresses the "Train-Serve Skew"in Global Horizontal Irradiance (GHI) forecasting through a proactive two-phase framework: evaluating model robustness via Monte Carlo noise injection and generating probabilistic intervals using Conformal Prediction. Our findings indicate that optimal model architectures depend heavily on local microclimates (MLP in Hanoi, LSTM in Son La, and XGBoost in Bac Giang). Under simulated weather uncertainty, all models degraded, with Bac Giang proving highly vulnerable due to its heavy reliance on dynamic meteorological features.

Crucially, Phase 2 demonstrated that relying solely on input noise yields dangerously low coverage (e.g., 7% in Son La). Applying Conformal Prediction successfully restored system reliability, elevating the Prediction Interval Coverage Probability (PICP) to over 85% across all noise scenarios while actively constraining the interval width (PIW). Practically, this framework transforms forecast uncertainty into quantifiable risk warnings, enabling safe power dispatch decisions for grid operators. Future work will validate this framework using ground-level SCADA data, explore advanced Conformal Prediction variants to further narrow the PIW under extreme noise, and integrate these forecasts directly into Battery Energy Storage System (BESS) optimization.

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